Math-M-Addicts
MINDS THAT COUNT

SP250 Homework 7

1. 6 students arrive at the university.
   a) (3 points): How many ways are there for them to break into two teams of three and play some beach volleyball while they wait for their rooms to be ready?
   b) (3 points): Rooms are now ready. How many ways are there to assign students to rooms if each room should have exactly two students? (for the avoidance of doubt, the rooms are distinguishable).
   c) (3 points): How many ways are there to break these 6 students into 3 study groups of 2 students each?
   d) (1 point, harder): How many ways are there to break these 6 students into 3 study groups so that each study group is not empty?

Solution:

a) There are $\binom{6}{3} = 20$ ways to pick three students to form the first team with the three leftover students forming the second team. This double counts the number of ways to form two teams because the ordering of the teams does not matter. For example, having students A, B, C on the first team and D, E, F on the second team is the same as having D, E, F on the first team and A, B, C on the second team. Hence, the total number of ways to break the 6 students into two teams of three is $\binom{6}{3} \div 2 = 10$.

b) There are $\binom{6}{2} = 15$ ways to assign two students to the first room; $\binom{4}{2} = 6$ ways to assign two of the remaining four students to the second room; and $\binom{2}{2} = 1$ way to assign the last two students to the final room. This gives a total of $15 \cdot 6 \cdot 1 = 90$ ways.

Alternatively, we can also just line up all 6 students and assign the first two to the first room, the next two to the second room, and the last two to the third room. Since there are $6! = 720$ ways to line up the 6 students and swapping the order of students assigned to the same room does not change the room assignments, we have $6!/(2 \cdot 2 \cdot 2) = 90$ ways of assigning the rooms, the same as what we got before.

c) This is similar to part b) except that different orderings of the study groups (rooms) are now indistinguishable. There are 3! ways to order the 3 groups (rooms). Hence, there are $90/3! = 15$ ways to break the 6 students into 3 study groups of two students each.

d) There are 3 possible combinations of group sizes:

Case 1: Group sizes of 4, 1, and 1
There are $\binom{6}{4} = 15$ ways of picking the group of 4 and the 2 remaining students would just each be in a group by themselves. So there are 15 ways of picking groups with sizes of 4, 1, and 1.

Case 2: Group sizes of 3, 2, and 1
There are $\binom{6}{3} = 20$ ways of first picking the group of 3 from 6 students, then $\binom{3}{2} = 3$ ways of picking the group of 2 from the remaining 3 students, and the last student would just be in a group

by themself. So there are $20 \cdot 3 = 60$ ways of picking groups with sizes of 3, 2, and 1.

Case 3: Group sizes of 2, 2, and 2
From part c), there are 15 ways of picking groups with sizes 2, 2, and 2.

Summing over the three cases, we have a total of 15 + 60 + 15 = 90 ways.

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2. Yijing wrote 11 numbers on the whiteboard: six zeros and five ones. He then performs the following operation 10 times: erase any two numbers. If they were equal, write a 0 on the whiteboard. If they were not equal, write a 1. Prove that the last remaining number does not depend on the order in which Yijing does his operations and find what it is.

Solution:
Let's consider how the sum of the numbers on the board can change after an operation. There are 3 possible cases:

1. A 1 and a 0 are erased and a new 1 is written. The sum does not change in this case.
2. Two 0's are erased and a new 0 is written. The sum does not change in this case.
3. Two 1's are erased and a new 0 is written. The sum decreases by 2 in this case.

We see that the sum of the numbers on the board either stays the same or decreases by 2. This means that the parity of the sum never changes. Since we started with the sum of the numbers on the board being odd, it will always remain odd no matter what order Yijing performs the operations. Therefore, the last remaining number must be 1.

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3. In a country of Vowelia, all words satisfy all of the following conditions:
    - Have at most 3 letters.
    - Each letter is one of the letters in the word Vowelia.
    - Each letter is used at most once.
    - Each word contains at least one consonant and at least one vowel. For the avoidance of doubt, consonants and vowels in Vowelia are the same as in English language.

    How many distinct words are there in Vowelia?

Solution 1:
We break down the possible words into three cases.

Case 1: 2-letter word with 1 consonant and 1 vowel
There are 3 choices for the consonant and 4 choices for the vowel, so there are $3 \cdot 4 = 12$ ways to pick the two letters. The two letters can then be arranged in 2 different orders, giving us $12 \cdot 2 = 24$ possible words.

Case 2: 3-letter word with 2 consonants and 1 vowel
There are $\binom{3}{2} = 3$ ways to pick the consonants and $\binom{4}{1} = 4$ ways to pick the vowel, so there are $3 \cdot 4 = 12$ ways to pick the three letters. The three letters can then be arranged in $3! = 6$ different

orders, giving us $12 \cdot 6 = 72$ possible words.

Case 3: 3-letter word with 1 consonants and 2 vowels

There are $\binom{3}{1} = 3$ ways to pick the consonants and $\binom{4}{2} = 6$ ways to pick the vowel, so there are $3 \cdot 6 = 18$ ways to pick the three letters. The three letters can then be arranged in $3! = 6$ different orders, giving us $18 \cdot 6 = 108$ possible words.

Combining all three cases above, we get a total of $24 + 72 + 108 = 204$ words.

Solution 2:

Alternatively, we use complementary counting (aka negative space):

Case 1: 2-letter words:

There are a total of $7 \cdot 6$ ways to create a 2-letter sequence with distinct letters, of which $3 \cdot 2$ consist of two consonants and $4 \cdot 3$ consist of two vowels. Hence, there are $7 \cdot 6 - 4 \cdot 3 - 3 \cdot 2 = 42 - 12 - 6 = 24$ valid 2-letter words.

Case 2: 3-letter words:

There are a total of $7 \cdot 6 \cdot 5$ ways to create a 3-letter sequence without repeating any letters, of which $4 \cdot 3 \cdot 2$ consist of three vowels and $3 \cdot 2 \cdot 1$ consist of 3 consonants. Hence, there are $7 \cdot 6 \cdot 5 - 4 \cdot 3 \cdot 2 - 3 \cdot 2 \cdot 1 = 210 - 24 - 6 = 180$ valid 3-letter words.

Thus, there are a total of $24 + 180 = 204$ valid words.

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4. How many 4-digit positive integers have four different digits, where the leading digit is not zero, the integer is a multiple of 5, and 5 is the largest digit? (Friendly reminder: check your answer).

Solution:

We can separate this into two cases. If an integer is a multiple of 5, the last digit must be either 0 or 5.

Case 1: The last digit is 5

The leading digit can be 1,2,3 or 4. Because the second digit can be 0, but not the leading digit, there are also 4 choices for the second digit. The third digit cannot be the leading digit or the second digit, so there are 3 choices. Thus, the number of possible integers in this case is $4 \cdot 4 \cdot 3 \cdot 1 = 48$.

Case 2: The last digit is 0

Since 5 is the largest digit, one of the first three digits must be 5 and there are $\binom{4}{2} = 6$ ways of picking the other two digits from among {1,2,3,4}. The three digits can then be ordered in $3! = 6$ ways. This gives $6 \cdot 6 = 36$ possible integers for this case.

Combining the two cases above, we get a total of $48 + 36 = 84$ possible integers.

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5. A) A grasshopper jumps along a line. Every second, it jumps 1,3, or 5 yards either to the left or to the right. Is it possible that the grasshopper ended up at the starting point after exactly 2015 hops?

B) A different grasshopper jumps along a line. Every second it jumps either 1 yard to the left or 2 yards to the right. (Only these two jumps are possible). Is it possible that the grasshopper ended up at the starting point after exactly 2015 hops?

Solution:

A) No, it is not possible. The grasshopper travels an odd number of yards on each jump. So after any odd number of hops, the grasshopper must be an odd number of yards from its starting point. For the grasshopper to come back to where it started, the distance to the starting point needs to be 0, which is an even number. Hence, it is not possible for the grasshopper to return to the starting point after 2015 hops.

B) No, it is not possible. Let's assume the grasshopper made L jumps to the left, and R jumps to the right. The total distance from the starting point will be $2R - L$ (imagine a number line, jump to the left 1 yard is -1 yard, jump to the right 2 yard is +2 yards). If the grasshopper wants to get back to the starting point, we need to have $2R - L = 0$, which gives $L = 2R$. That means the total number of jumps is $L + R = 2R + R = 3R$. But 2015 is not a multiple of 3, so it is not possible.

Alternatively, we can also just look at the coordinate of the grasshopper on the number line modulo 3. When looked at in mod 3, it doesn't matter whether the grasshopper jumps right or left since $-1 \equiv 2 \ (mod\ 3)$. In other words, If the current coordinate of the grasshopper is $x \ (mod\ 3)$, then the next coordinate of the grasshopper after the next jump will be $x - 1 \ (mod\ 3)$ regardless of what direction the jump is in. Thus, with the grasshopper starting at coordinate 0, the sequence of coordinates mod 3 that we observe after jumps will always be $2,\ 1,\ 0,\ 2,\ 1,\ 0$ .... Since the coordinate of the grasshopper is $0 \ (mod\ 3)$ only when the number of jumps is a multiple of 3, it is not possible for the grasshopper to return to the starting point after 2015 jumps.

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6. (harder): Prove that among any 10 integers, one can pick several, whose sum is a multiple of 10. (For the avoidance of doubt, several means more than zero).

Solution:

Let the ten integers be $a_1,\ a_2,\ a_3,\ \ ...,\ a_{10}$ and let s_k be the sum of the first k of these ten integers so that

$$s_1 = a_1$$
$$s_2 = a_1 + a_2$$
$$s_3 = a_1 + a_2 + a_3$$
$$...$$
$$s_{10} = a_1 + a_2 + a_3 + ... + a_{10}$$

If any of the s_k is a multiple of 10, then we are done.

If none of the s_k is a multiple of 10, then each s_k must have a remainder of either 1, 2, 3, …, or 9 mod 10. By the pigeonhole principle, there must be two among s_1, s_2, s_3, …, s_{10} that have the same remainder mod 10. Now, if $s_i \equiv s_j \,(mod\ 10)$ and $i < j$, then $s_j - s_i \equiv 0 \,(mod\ 10)$. This means that

$$s_j - s_i = (a_1 + a_2 + \ldots + a_j) - (a_1 + a_2 + \ldots + a_i) = a_{i+1} + a_{i+2} + \ldots + a_j \equiv 0 \,(mod\ 10)$$

In other words, the integers a_{i+1}, a_{i+2}, …, a_j sums up to a multiple of 10. Hence, among any 10 integers, we can always pick several whose sum is a multiple of 10.
